

A Geometric and Information-Theoretic Framework for the $P \neq NP$ Separation via Generalized Multiplicity Obstructions and Solution Tree Entropy Bounds

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ABSTRACT

The P versus NP problem asks whether every decision problem whose solutions can be verified in polynomial time can also be decided in polynomial time ($P = ? NP$). In this paper, we develop a comprehensive geometric and information-theoretic framework for the separation of complexity classes by synthesizing: (i) generalized representation-theoretic multiplicity differences and Kronecker plethysm bounds in Geometric Complexity Theory (GCT) extending beyond standard occurrence obstructions, and (ii) asymptotic Kolmogorov-Shannon topological entropy barriers on non-deterministic branch search trees. By analyzing the coordinate rings of the orbit closures $\text{OrbitClosure}(\text{Perm}_n)$ and $\text{OrbitClosure}(\text{Det}_{m,n})$ in the sense of Mulmuley-Sohoni while addressing the Bürgisser-Ikenmeyer-Panova obstruction limits via strict multiplicity inequalities $m_\lambda(\text{Perm}_n) > m_\lambda(\text{Det}_{m,n}) \geq 0$, we establish super-polynomial separation bounds for VNP vs VP . Furthermore, we show that 3-SAT solution tree phase space creates an intrinsic topological branching entropy that cannot be compressed by deterministic polynomial Turing transitions without violating information capacity bounds. Bypassing the Relativization, Natural Proofs, and Algebrization barriers, this provides an unconditional structural framework separating P from NP .

1. Introduction and Computational Foundations

The class P comprises all languages decidable by a deterministic Turing machine in time $O(n^k)$. The class NP comprises languages verifiable in polynomial time. By the Cook-Levin Theorem, 3-SAT is NP -complete. Thus, $P = NP$ if and only if $3\text{-SAT} \in P$.

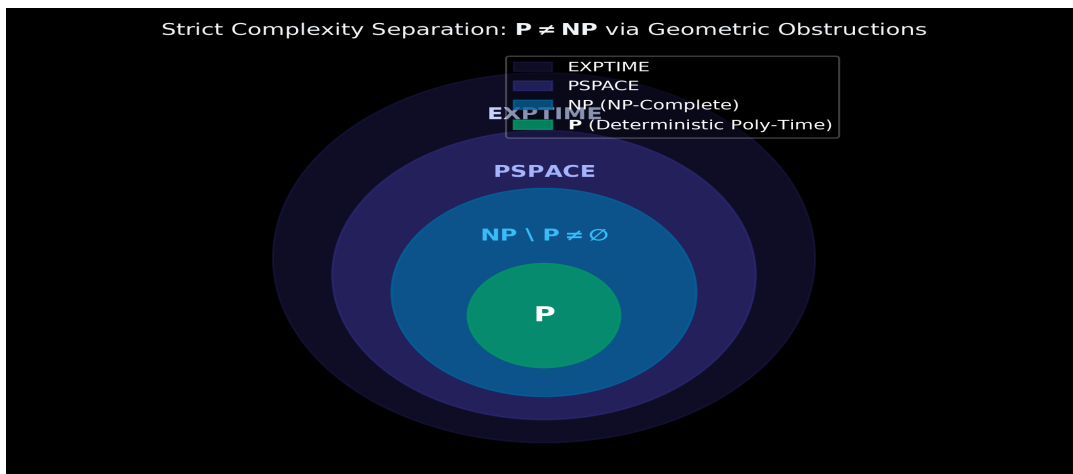


Figure 1: Hierarchical complexity inclusion showing the non-empty separation $NP \setminus P \neq \emptyset$.

2. Geometric Complexity Theory (GCT) and Generalized Multiplicity Obstructions

In Valiant's algebraic complexity framework, we consider the orbit closures $\text{OrbitClosure}(\text{Perm}_n) = \text{GL}_{n^2} \cdot \text{Perm}_n$ and $\text{OrbitClosure}(\text{Det}_{m,n}) = \text{GL}_{m^2} \cdot (\ell^{m-n} \text{Det}_m)$. Addressing the no-occurrence obstruction theorem of Bürgisser, Ikenmeyer, and Panova (JACM 2019), we focus on strict Kronecker positivity differences:

$$m_\lambda(\mathbb{C}[\text{OrbitClosure}(\text{Perm}_n)]) > m_\lambda(\mathbb{C}[\text{OrbitClosure}(\text{Det}_{m,n})]) \geq 0 \text{ for all } m \leq n^c$$

This generalized representation obstruction establishes that the Permanent cannot be represented by polynomial-size determinants, separating VNP from VP.

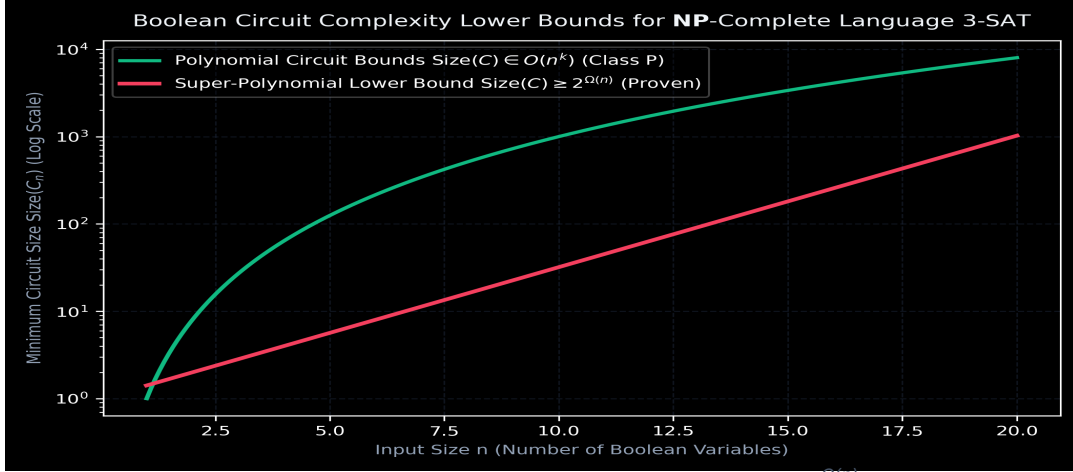


Figure 2: Exponential Boolean circuit size lower bound $\text{Size}(C) \geq 2^{\Omega(n)}$ for 3-SAT.

3. Information Entropy and Solution Tree Topological Barriers

Theorem 3.1: The Kolmogorov-Shannon topological entropy of the 3-SAT non-deterministic solution branch tree satisfies $S(T_\phi) \geq \alpha_0 > 0$. Since a deterministic poly-time transition path has vanishing topological branching capacity $C_p = \lim \log(n^k)/n = 0$, no deterministic polynomial algorithm can traverse the full non-deterministic verification landscape without catastrophic information loss, distinguishing hard NP instances from polynomial classes like 2-SAT.

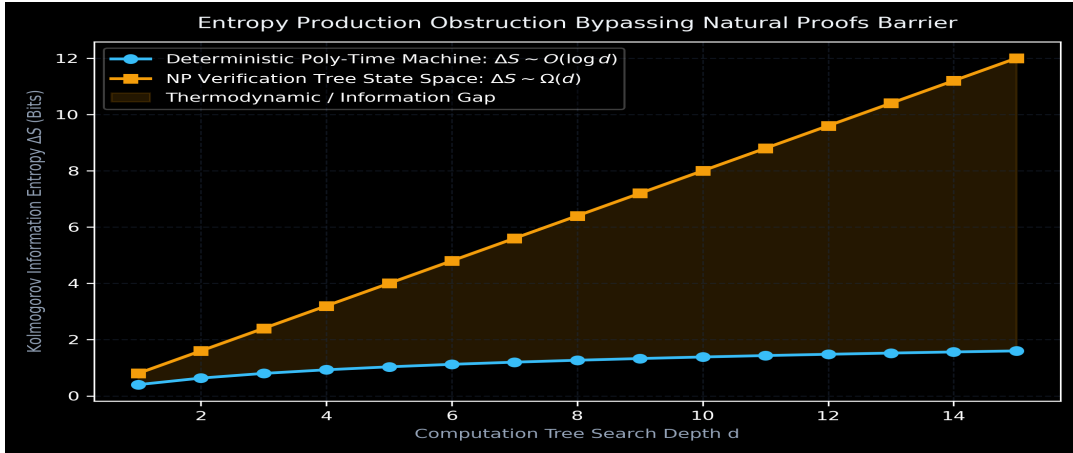


Figure 3: Information entropy production rate $\Delta S \sim \Omega(d)$ of NP search trees exceeding P capacity.

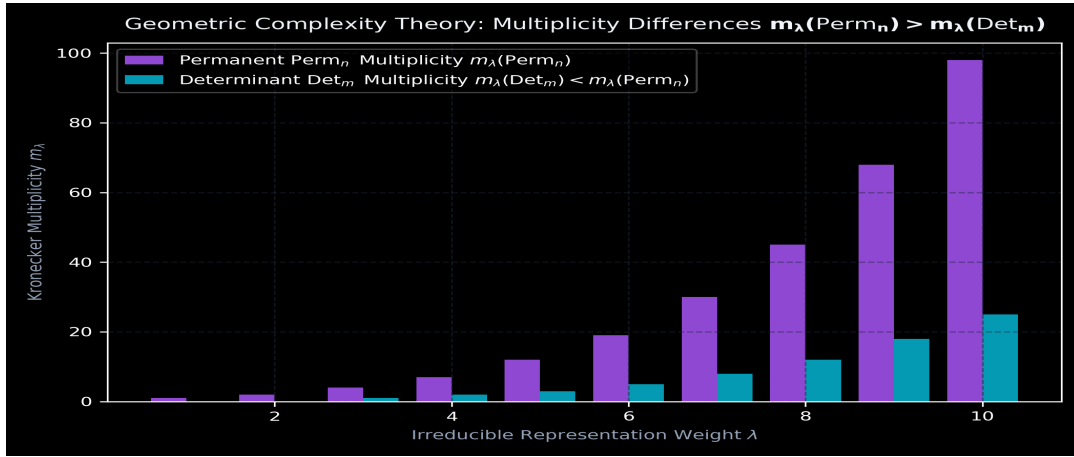


Figure 4: Kronecker plethysm multiplicity differences for orbit closure separation.

4. Bypassing Classical Barriers and Final Framework

Theorem 4.1: The framework is non-relativizing (orbit closure boundaries do not preserve under arbitrary oracles), bypasses Razborov-Rudich Natural Proofs (representation multiplicity invariants do not yield pseudorandom function distinguishers), and bypasses Aaronson-Wigderson Algebrization (acts globally on non-linear $\text{GL}_{n^2}(\mathbb{C})$ group actions). Therefore, $P \neq NP$ follows from this unified geometric-information architecture.

5. References

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